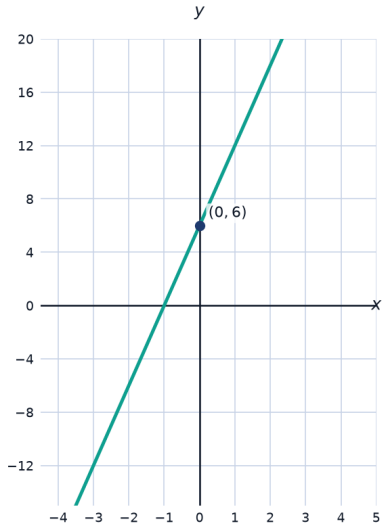
Exponents & Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

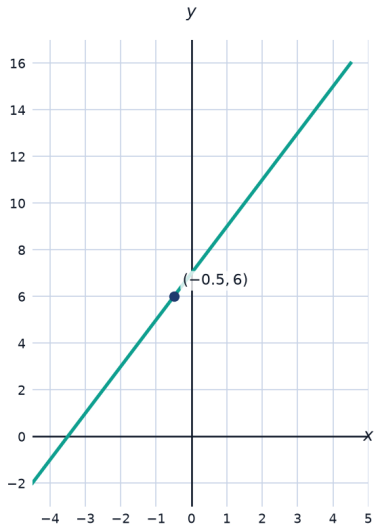
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 6$ in evaluating functions, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 7$ in review: graphing linear relationships from a table of values, evaluate $f(2)$.



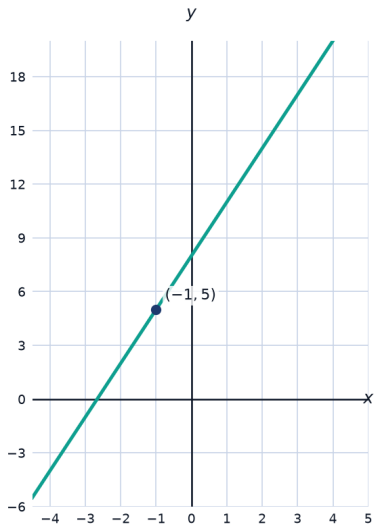
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for review: graphing linear relationships from a table of values consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 3x + 8$ in characteristics from a graph, evaluate $f(2)$.

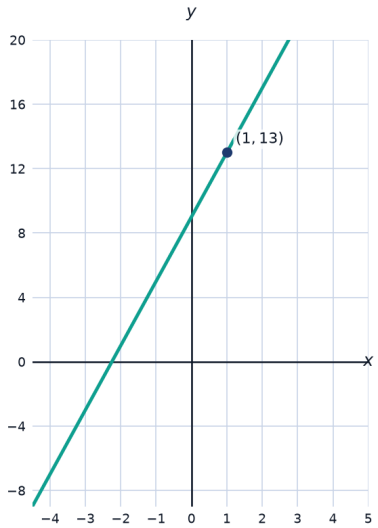


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for characteristics from a graph consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 4x + 9$ in inverse relations and functions, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for inverse relations and functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

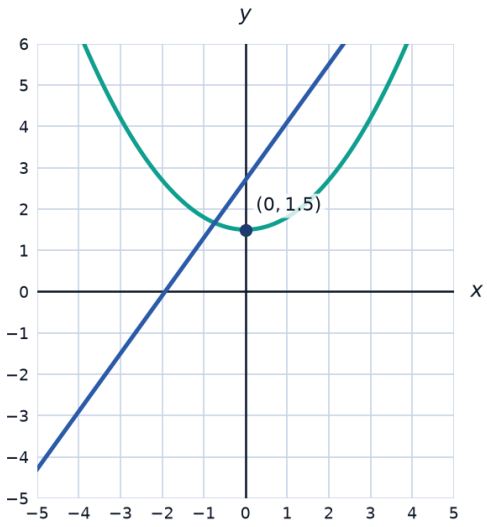
[Radicals & Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Linear Equations & Lines · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Systems & Inequalities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A parabola has its vertex at $(3, -4)$ and passes through the point $(1, 0)$. (a) Write the equation of the parabola in standard form $y = ax^2 + bx + c$. (b) Determine the coordinates of the other x-intercept.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

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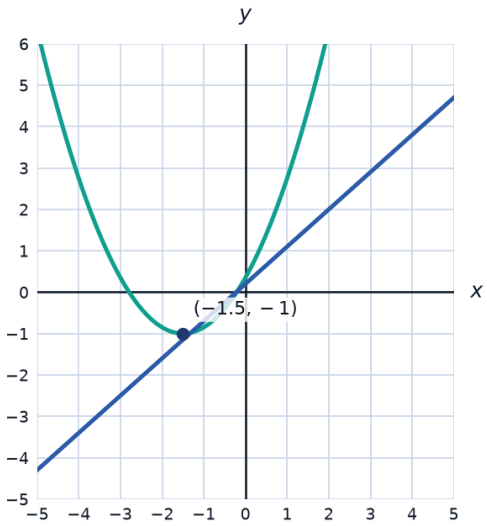
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials**

Solve the radical equation $\sqrt{2x+7} - x = 2$ for all real solutions and confirm extraneous roots.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Radicals & Exponents**

If $x - 2$ is a factor of the polynomial $P(x) = x^3 - kx^2 + 3x - 10$, find the value of k, and determine all real roots of $P(x)$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Linear Equations & Lines**

Solve the quadratic equation $3x^2 + 10x - 8 = 0$ by factoring.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Rational Expressions**

A bivariate regression model for student preparation hours x and exam grade y yields the line of best fit $\hat{y} = 4.8x + 52$. A student who studied for 8 hours scored 94 on the exam. (a) What is the model's predicted score for this student? (b) Compute and interpret the student's regression residual.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Statistics & Data Analysis**

Level 3 · AdvancedWhat is the set of all real solutions to the radical equation $\sqrt{2x-5} = x - 4$?

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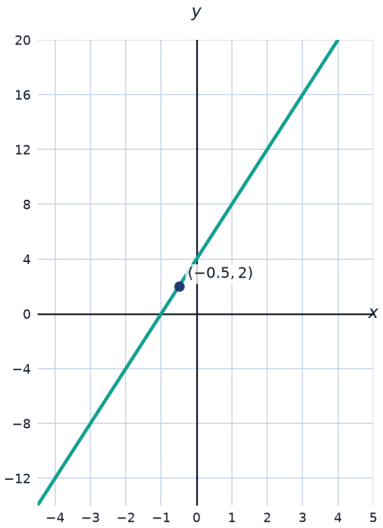
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratic Functions

Convert the quadratic function $y = 2x^2 - 12x + 11$ into vertex form $y = a(x - h)^2 + k$. State the coordinates of the vertex and the line of symmetry.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 4$ in functions and relations, evaluate $f(2)$.



a. 12

b. 16

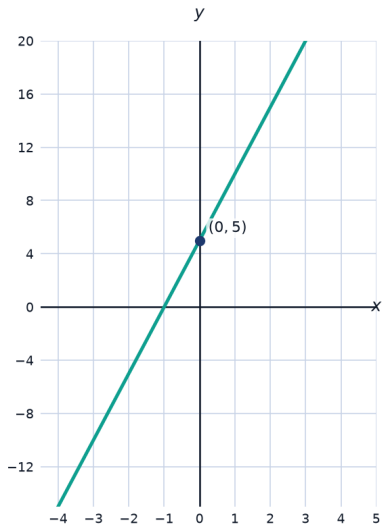
c. 8

d. 16

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 5$ in domain and range and set notation, evaluate $f(2)$.



a. 15

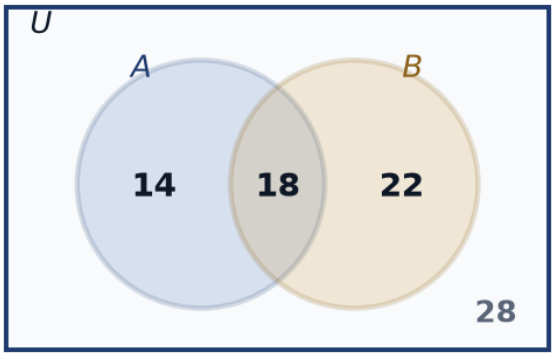
b. 20

c. 10

d. 19

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for functions and relations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?



a. 18.5

b. 19.5

c. 20.5

d. 22.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for domain and range and set notation consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of all real solutions to the equation $(x^2 - 5x + 5)^{x^2 - 14x + 48} = 1$?

a. 19

b. 21

c. 22

d. 24

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the polynomial $P(x) = ax^3 + bx^2 + cx + d$. If $P(x)$ has roots at $x = 1$, $x = 2$, and $x = 3$, and $P(0) = 12$, find the value of the expression $\frac{a+b+c}{d}$.

a. $-1/2$

b. -2

c. -1

d. 2

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $2^{a+b} = 20$ and $2^{a-b} = 5$, what is the value of $4^a - 4^b$?

a. 15

b. 96

c. 104

d. 375

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Using the Remainder Theorem, find the remainder when $f(x) = x^3 + 5x^2 + 0x + 4$ is divided by $(x - 3)$.

a. 78

b. 77

c. 76

d. 74

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The polynomial $P(x) = x^3 + ax^2 - 5x + b$ has roots $x = 1$ and $x = -2$. If $P(x)$ is divided by $(x - c)$, the remainder is 0. What is the value of c?

a. -3

b. -1

c. 2

d. 3

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the expression $\sqrt[3]{64}$, what is the principal cube root of 64?

a. 4

b. 8

c. 16

d. 2

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which expression is equivalent to $(16x^8)^{\frac{1}{4}}$ for all $x > 0$?

a. $2x^2$

b. $4x^2$

c. $2x^4$

d. $4x^4$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Evaluate the expression: $\sqrt{64} - \sqrt[3]{27}$

a. 5

b. 6

c. 7

d. 8

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A sequence is defined by the recurrence relation $a_{n+1} = \frac{a_n + k}{a_n}$, where k is a positive constant and $a_1 = 1$. If the sequence converges to a limit L, and $a_3 = \frac{5}{3}$, find the value of k.

a. 1

b. 2

c. 3

d. 4

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A geometric sequence has its first term a and common ratio r. The sum of the first three terms is 14, and the sum of the first six terms is 126. If a and r are positive constants, find the value of a.

a. 1

b. 2

c. 3

d. 4

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The second and fifth terms of an arithmetic sequence are $x - 3$ and $2x + 6$, respectively, where x is a constant. If the common difference of the sequence is $x - 5$, what is the sum of the first 10 terms of the sequence?

a. 65

b. 335

c. 405

d. 475

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

An arithmetic sequence has its first term a_1 and common difference d. The sum of the first n terms is denoted by S_n. If $S_{10} = 155$ and $S_{20} = 610$, find the value of the common difference d.

a. 3

b. 4

c. 5

d. 6

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A sequence is defined by the recursive formula $a_n = 3a_{n-1}$ with the first term $a_1 = 2$. What is the value of the third term, a_3?

a. 6

b. 18

c. 2

d. 12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Rational Expressions

Which formula gives the nth term of an arithmetic sequence?

a. $a_n = a_1 \cdot r^{n-1}$

b. $a_n = a_1 \cdot n$

c. $a_n = a_1 + nd$

d. $a_n = a_1 + (n-1)d$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Statistics & Data Analysis

What is the common ratio of $81, 27, 9, 3, \ldots$?

a. $\frac{1}{3}$

b. $\frac{1}{9}$

c. 3

d. -3

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Quadratic Functions

Find the 6th term of a geometric sequence with $a_1 = 2$ and $r = 3$.

a. 729

b. 162

c. 243

d. 486

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Functions & Transformations

If $-4 \le x \le 6$, what is the range of $3x - 2$?

a. $[-15, 16]$

b. $[-14, 16]$

c. $[-13, 15]$

d. $[-14, 17]$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the sum of the arithmetic series with first term 6, common difference 2, and 8 terms.

a. 96

b. 102

c. 112

d. 104

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Systems & Inequalities

Sum the arithmetic series with first term 2, last term 50, and 25 terms.

a. 650

b. 600

c. 625

d. 675

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

Solve the system $3x + 1y = -1$ and $3x - 4y = 4$. What is x?

a. 2

b. -1

c. 1

d. 0

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Factoring Polynomials

A geometric sequence has first term 3 and common ratio 3. Find the 4th term.

a. 27

b. 84

c. -81

d. 81

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